

Drew Council

770-833-1759 | NYC | andrew.p.council@gmail.com | github.com/AndrewCouncil | linkedin.com/in/drew-council | drewcouncil.com

Education

Duke University

Bachelor of Computer Science, Electrical & Computer Engineering

Durham, NC

Graduating 2024

- Completed classes in Data Structures/Algorithms, Computer Architecture, Networking, and Robotics.

Skills

Python, C, Linux, ROS, ROS2, Docker, Git, Bash, Arduino, Verilog, Raspberry Pi, KiCad, AWS, C++, Java, Rust

Experience

Software Engineer

Jan 2022 — Present

BotBuilt Robotics

Durham, NC

- Worked with others on an agile team to develop software for robotic construction of houses.
- Headed company DevOps, including managing CI/CD, developer tools, and embedded software deployment.
- Designed embedded hardware for a ROS2 network, providing services for actuator and sensor control.
- Containerized ROS2 applications into Docker images for several platforms and hardware configurations.
- Spearheaded CI for testing and development including automatic Docker container testing on pull requests.
- Integrated AWS hosting for automated Docker image builds with custom logging and error reporting.

Software Subteam Lead

Jun 2020 — Jun 2022

Duke University Robotics Club

Durham, NC

- Competed in annual [RoboSub robot competition](#), where we designed a fully autonomous submarine robot to complete a variety of complex maneuvering and manipulation tasks in an unfamiliar underwater environment.
- Coordinated a 25+ member agile environment team using ROS, Docker, and Git to manage a shared codebase.
- Implemented PID, Sensor Fusion, Computer Vision, and SMACH to improve robot accuracy and capability.
- Earned 1st in Propulsion System, 3rd in Sensor optimization in 2021; 1st in technical report in 2021 and 2022.

Teaching Assistant

Aug 2021 — Dec 2021

Duke First Year Engineering Design

Durham, NC

- Guided small groups of students in project management throughout a semester-long design challenge.
- Provided technical expertise in actuators, programming embedded systems, sensors, and PCB design.

Projects

Duke General Robotics Lab, Cell Robots Research

Feb 2023 — Present

- Designed omnidirectional grounded robot swarm for research in distributed control and communication.
- Researched algorithms to control a swarm of robots to perform tasks with low communication and orchestration.

Duke Product Design, "Aelevate" Bike Trainer

Sep 2022 — Dec 2022

- Used PlatformIO to write C++ embedded software to read sensors and control motors on a tilting bike trainer.
- Created a serial interface to communicate with a custom GUI application for touchscreen user control.

Duke Computer Engineering, FPGA Typeracer-style Arcade Game

Sep 2022 — Dec 2022

- Used Verilog to implement a MIPS-like pipelined processor to process serial input and generate VGA graphics.
- Implemented a head-to-head typing arcade game including display and keyboard drivers on an FPGA.

Duke Engineering First Year Design, Room Availability Detection System

Sep 2020 — May 2021

- Used a Raspberry Pi and Python to analyze and error correct PIR sensor data and detect a person's presence.
- Communicated this sensor data over a custom web API to publish room status and use statistics to a website.